DS5899: Special Topics in Data Science
Leveraging NLP in Asset Management

Week 8: Prompting

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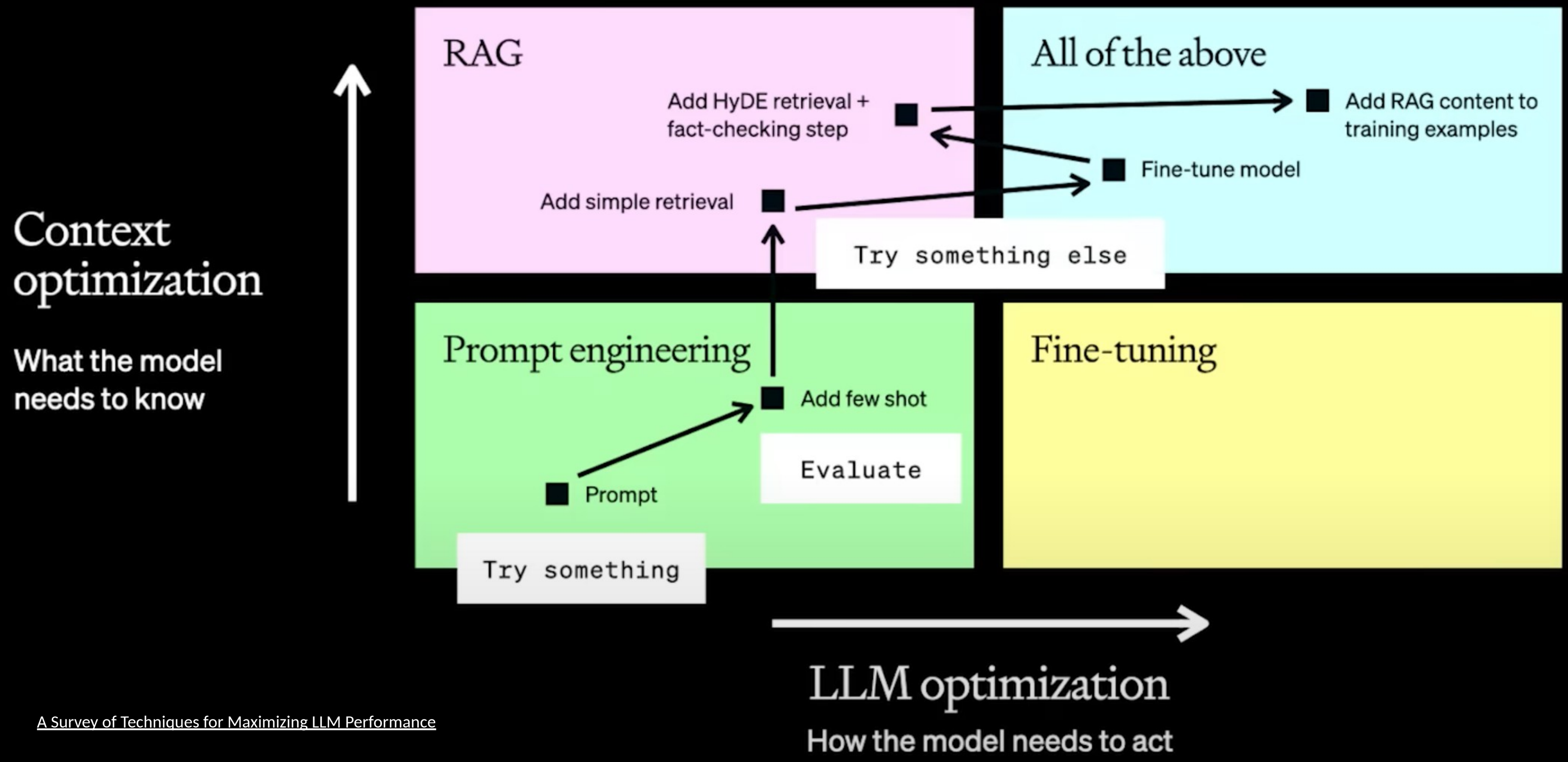
The optimization flow



A Survey of Techniques for Maximizing LLM Performance

Part 1:
Prompt Engineering

LLMs Are Gettering Longer Contexts

Proprietary Models	Context Window	Notable Capability
Google Gemini 3 Pro	**1M - 2M tokens**	The gold standard for multi-modal context (video/audio).
Claude 4.5 Sonnet	**1M tokens**	Exceptional at "agentic" tasks like coding across entire repos.
OpenAI GPT-5.2	**400K tokens**	Focused on high-reasoning accuracy over raw length.
Claude 4.6 Opus	**1M tokens**	Features "Infinite Chats" to mitigate context errors.
Grok 4.1 Fast	**2M tokens**	Optimized for real-time data ingestion from X (Twitter).

Open Source Models	Context Window	Best Use Case
Meta Llama 4 Scout	**10M tokens**	Currently holds the record; designed to replace RAG.
Mistral Large 3	**256K tokens**	Massive 675B MoE model; very efficient for its size.
Qwen 3-Coder	**256K - 1M***	Specialized for repository-scale code understanding.
DeepSeek-R1	**164K tokens**	A reasoning-first model with high "needle-in-a-haystack" recall.
Kimi K2.5	**256K tokens**	Popular for high-speed document reasoning.

Lost in the Middle: How Language Models Use Long Contexts

- Changing the location of relevant information within the language model's input context results in a U-shaped performance curve.

Input Context

Write a high-quality answer for the given question using only the provided search results (some of which might be irrelevant).

Document [1](Title: Asian Americans in science and technology) Prize in physics for discovery of the subatomic particle J/ψ. Subrahmanyan Chandrasekhar shared...
Document [2](Title: List of Nobel laureates in Physics) The first Nobel Prize in Physics was awarded in 1901 to Wilhelm Conrad Röntgen, of Germany, who received...
Document [3](Title: Scientist) and pursued through a unique method, was essentially in place. Ramón y Cajal won the Nobel Prize in 1906 for his remarkable...

Question: who got the first nobel prize in physics
Answer:

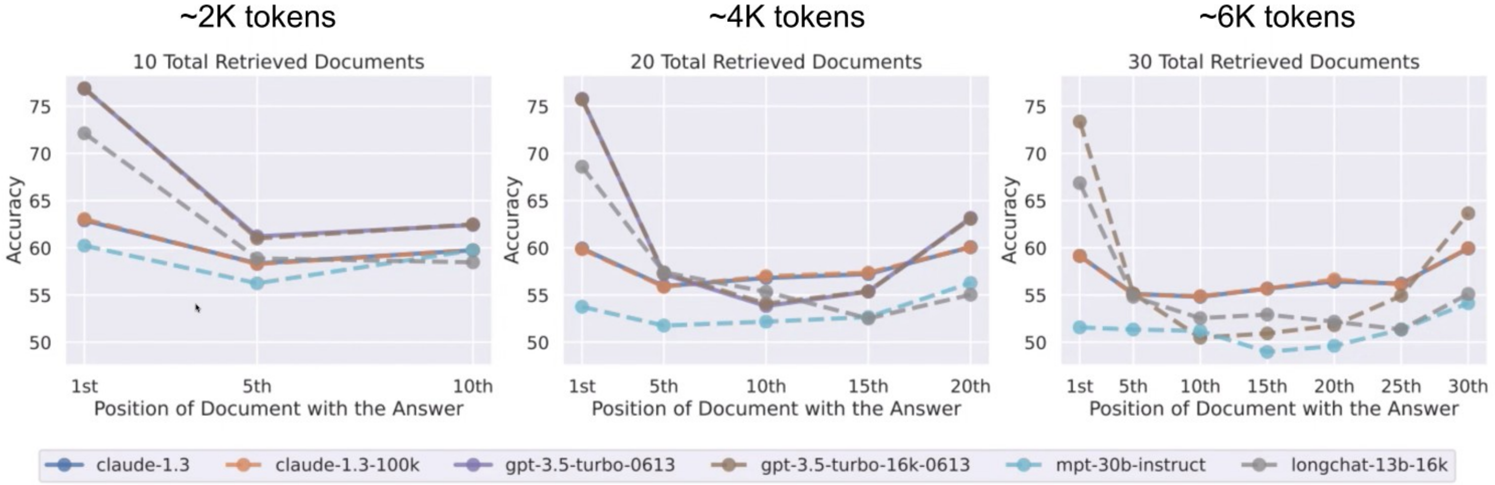
Desired Answer

Wilhelm Conrad Röntgen

Figure 2: Example of the multi-document question answering task, with an input context and the desired model answer. The document containing the answer is bolded within the input context here for clarity.

A U-shaped Performance Curve

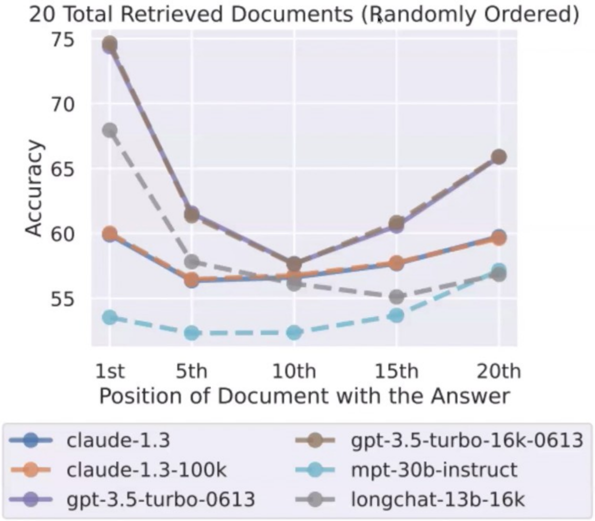
- Models are particularly good at using information at the beginning (primacy bias) or the end (recency bias) of their input context. However, their performance tends to decline when they need to recall and apply information located in the middle of the input context.

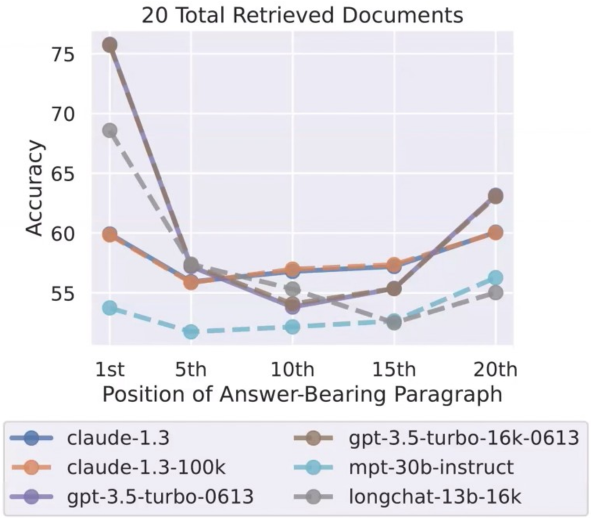


Best Closed-Book performance: GPT-3.5-Turbo, ~56%
Best Oracle (only feed in relevant doc) performance: GPT-3.5-Turbo, ~88.5%

Is It Due to the Prompt Mention “Search Results”?

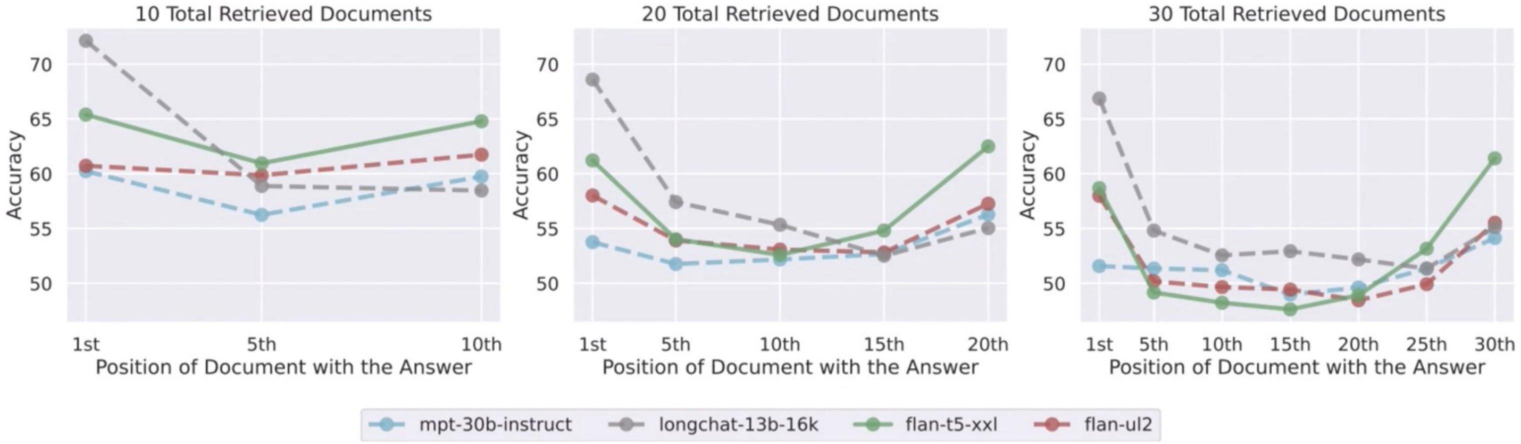
- To see if model could be "forced" to pay equal attention to the entire context, instructions are changed:
  - **Original Prompt:** Simply asked the model to use the search results.
  - **Modified Prompt:** Explicitly told the model that the results were **ordered randomly** and that some might be **irrelevant**.





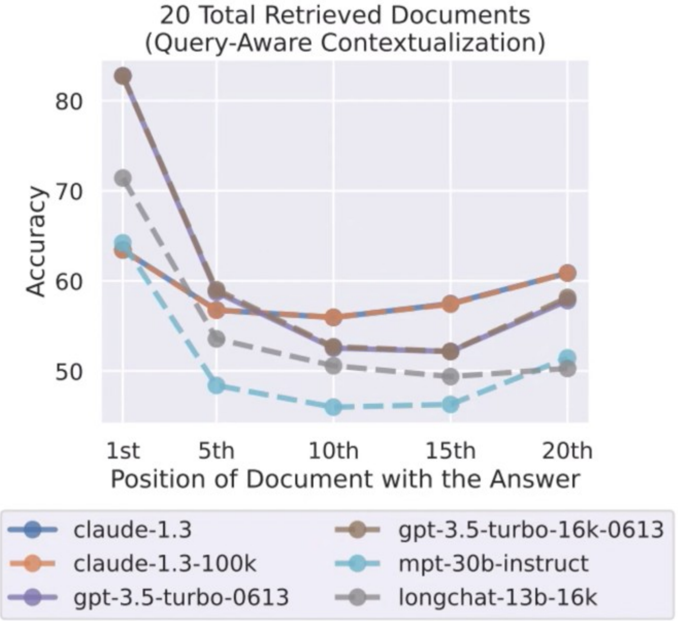
Is It Due to the Effect of Decoder Model Architecture?

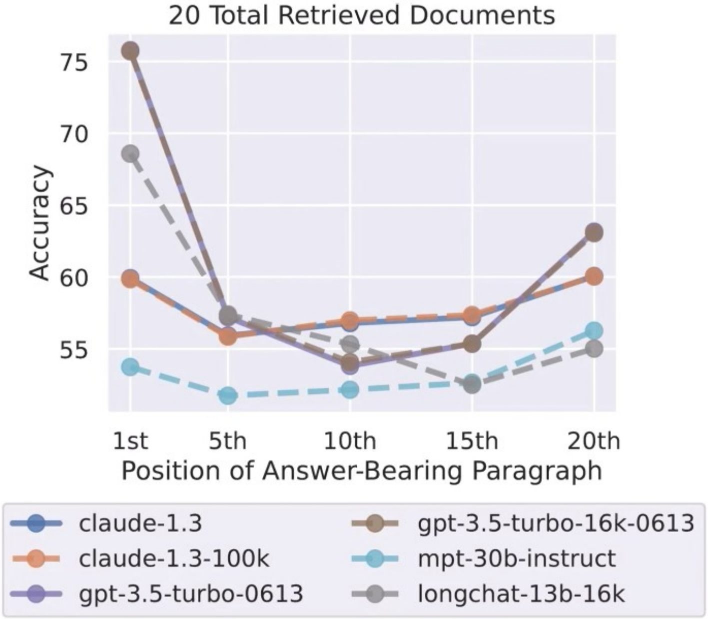
- When the total amount of information provided to the model is within **2048 tokens**, the model is operating within its "comfort zone".
- Once a prompt has longer than 2048 tokens, the model's performance begins to degrade in a very specific pattern known as the U-shaped curve.
- ‘Encoder-Decoder’ models (like Flan-UL2 and Flan-T5-XXL) actually perform slightly better than "Decoder-only" models on longer sequences, but they still eventually succumb to the U-shaped degradation.



Does the Placement of the Query at the End of the Context Have an Impact on the Result?

- Since decoder-only models don't see the question until the very end of the context, their representations of document tokens are query-independent
- What if we put the query before *and* after the documents?

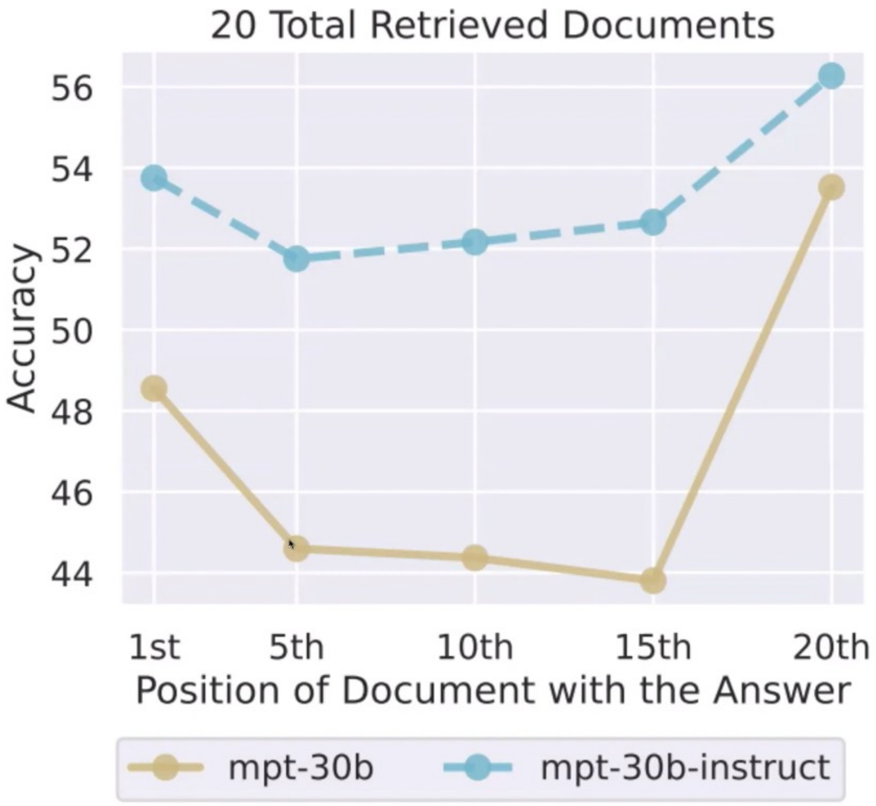




Is It Due to Effect of Instruction Fine-tuning?

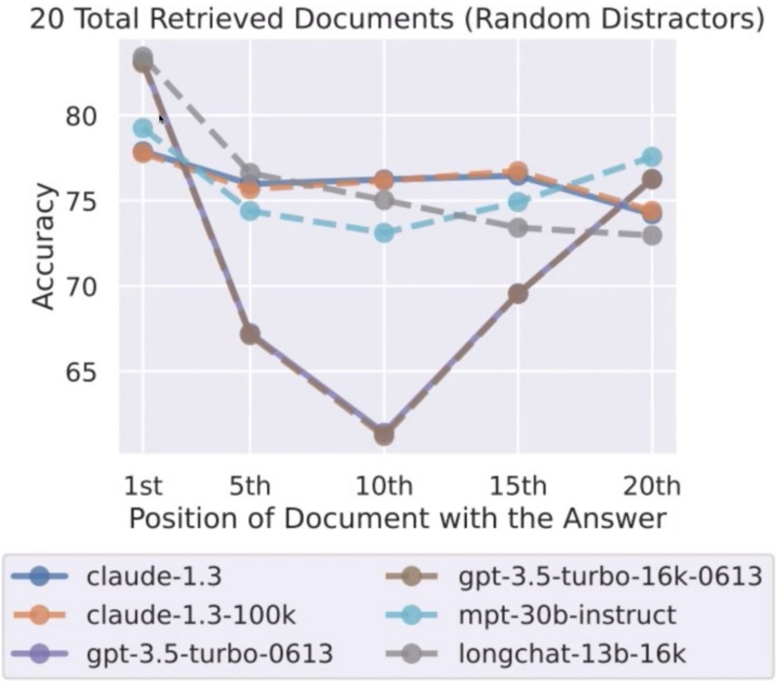
- Intuition: instruction fine-tuning generally places instruction at very beginning or very end, which may cause this U-shaped perf.

- Surprisingly, that doesn't seem to be in the case.



Is It Because LMs Get Distracted by Retrieved Documents?

- What happens if you use random distractors instead of retrieved distractors?
- Surprisingly, model performance still sees a sizable degradation.



- Random Distractors: documents chosen completely at random from a large dataset.
- Retrieved Distractors: documents topically related to the user's question but do not actually contain the answer.

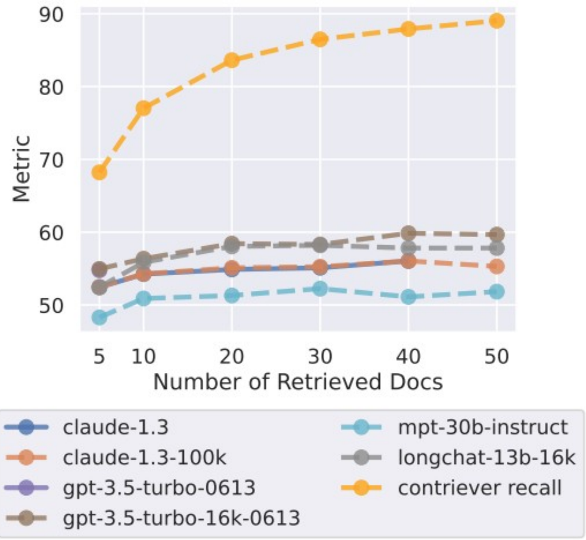
In Summary

- **Is this due to the prompt mentioning “search results”?**
  - No, modified prompt still shows degradation
- **Is this specific to decoder-only LMs?**
  - Encoder-decoder slightly better but still degrades on longer contexts.
- **Is this because we put the query only at the end of the context?**
  - No, putting query at start and end still results in degradation
- **Is this a byproduct of instruction-tuning?**
  - No, non-instruction tuned models also show these trends.
- **Is this because LMs get distracted by retrieved documents (hard negatives*)?**
  - No, performance degrades also on random negatives

* In information retrieval, a "hard negative" refers to a document that is considered irrelevant to a given query but is very similar in content or features to a relevant document, making it difficult for a retrieval model to accurately distinguish between the two.

Is More Context Always Better?

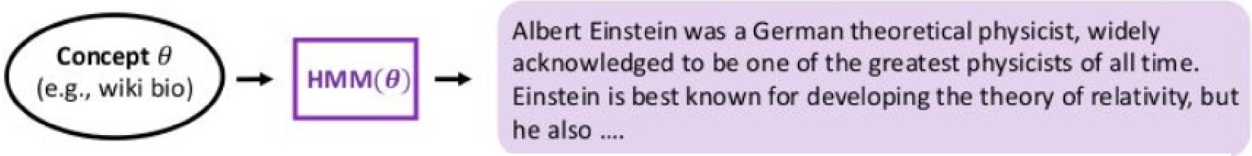
- As the number of retrieved documents increases, model performance saturates long before retriever recall, indicating that the models have difficulty making use of the extra retrieved documents.



***Contriever recall** can be interpreted as the probability that a relevant document will be retrieved by the model.

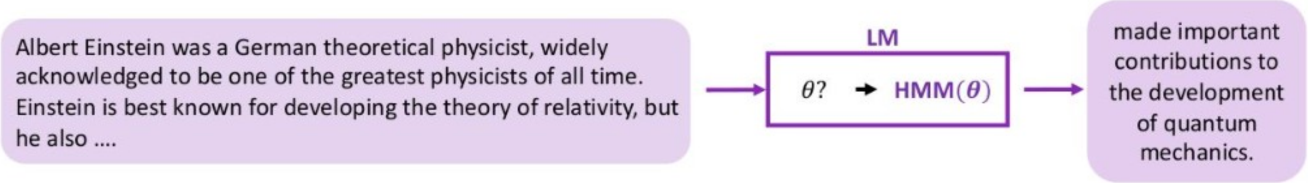
Pretraining distribution: mixture of HMMs (Hidden Markov Models)

Concept θ encodes transitions in HMM



Intuition

Language model **implicitly** infers latent concept θ

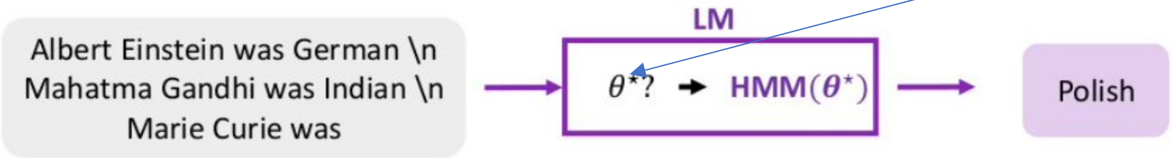


Bayesian inference view of in-context learning

Intuition

Can the language model infer concept θ^*?

all the components of the prompt (input distribution, the output space and format) are providing “evidence” to enable the model to better infer (locate) concepts that are learned during pretraining.



Difficulty: condition on samples from prompt distribution, not training distribution!

[Source: Xie at al., An Explanation of In-context Learning as Implicit Bayesian Inference, July 2022]

Bayesian inference view of in-context learning, Continued

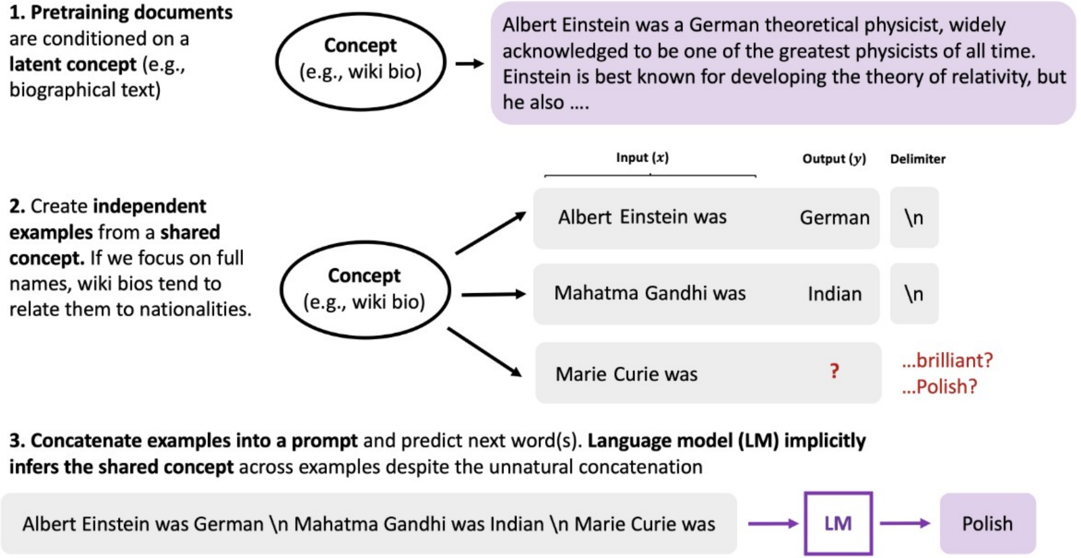


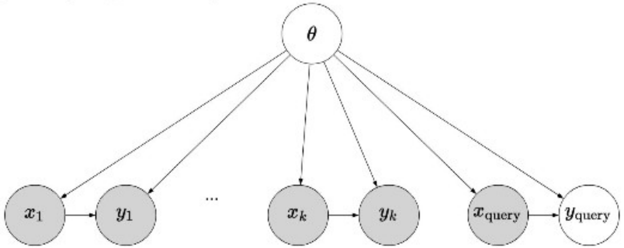
Figure 1: In-context learning can emerge from modeling long-range coherence in the pretraining data. During pretraining, the language model (LM) implicitly learns to infer a latent concept (e.g., wiki bios, which typically transition between name (Albert Einstein) → nationality (German) → occupation (physicist) → ...) shared across sentences in a document. Although prompts are unnatural sequences that concatenate independent examples, in-context learning occurs if the LM can still infer the shared concept across examples to do the task (name → nationality, which is part of wiki bios).

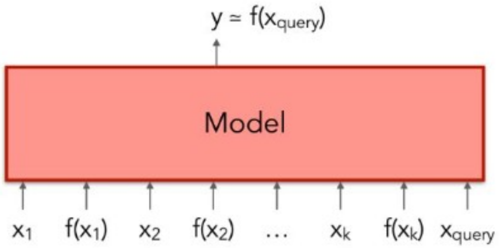
[Source: Xie at al., An Explanation of In-context Learning as Implicit Bayesian Inference, July 2022]

In-context Learning as Implicit Bayesian Inference

Bayesian inference

Posit a latent concept θ (e.g., task)





$$\underbrace{p(y_{\text{query}} \mid x_1, y_1, \ldots, x_k, y_k, x_{\text{query}})}_{\text{posterior predictive}} = \int p(y_{\text{query}} \mid x_{\text{query}}, \theta) \underbrace{p(\theta \mid x_1, y_1, \ldots, x_k, y_k)}_{\text{posterior}} d\theta$$

[Source: Xie at al., An Explanation of In-context Learning as Implicit Bayesian Inference, July 2022]

Transformer directly fits the posterior predictive distribution!

$$p(\text{output}|\text{prompt}) = \int_{\text{concept}} p(\text{output}|\text{concept}, \text{prompt}) p(\text{concept}|\text{prompt}) d(\text{concept})$$

- Ideally, p(concept|prompt) concentrates on the prompt concept with more examples in the prompt so that the prompt concept is “selected” through marginalization.

Prompts provide noisy evidence for Bayesian inference

- The transitions within training examples (green in the figure) allow the LM to infer the latent concept they all share. In a prompt, the green transitions come from below, which all provide signal for Bayesian inference.
  - the input distribution (the transitions inside the news sentence),
  - the output distribution (the topic word),
  - the format (syntax of news sentence),
  - the input-output mapping (relation between the news and the topic) Because the training examples are IID, concatenating them together often creates unnatural, low-probability transitions between examples (red in the figure) .
- if the signal is greater than the noise, then the LM can successfully do in-context learning:

Circulation revenue has increased by 5% in Finland. // Finance

They defeated … in the NFC Championship Game. // Sports

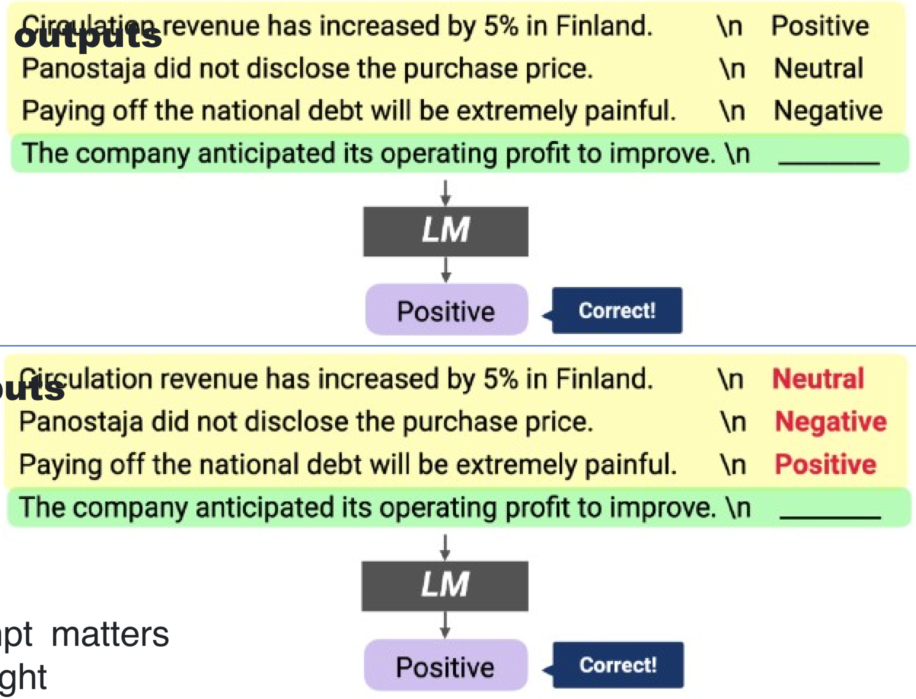
Apple … development of in-house chips. // Tech

[How does in-context learning work? A framework for understanding the differences from traditional supervised learning]

Which aspects of the prompt are most important for in-context learning?

Examples with ground truth outputs

Examples with random outputs

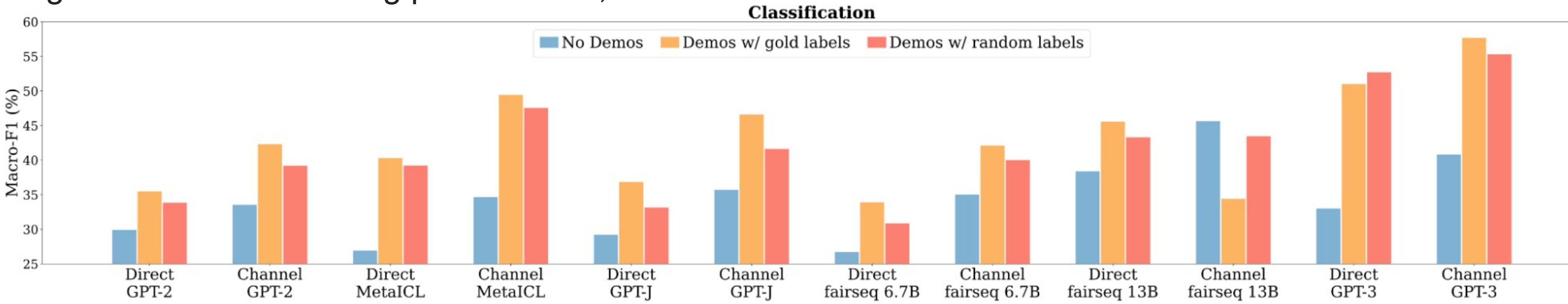


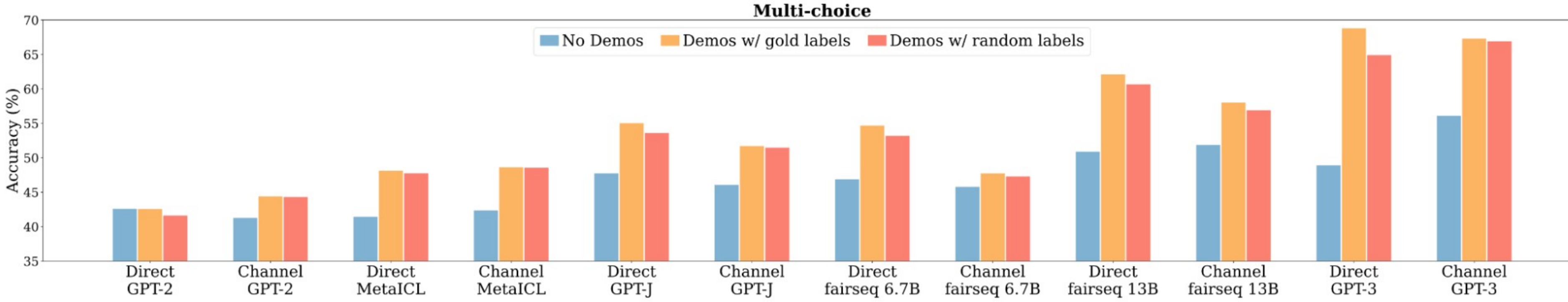
- Input-output pairing in the prompt matters much less than previously thought

The prompt with ground truth outputs (top) and the prompt with random outputs (bottom).

[How does in-context learning work? A framework for understanding the differences from traditional supervised learning]

- Aphthous experiment with 12 models whose sizes range 774M - 175B, including the largest GPT-3.
- Replacing ground truth outputs with random outputs hurts performance significantly less than previously thought and is still significantly better than no-examples.
- This means, unlike typical supervised learning, ground truth outputs are not really required to achieve good in-context learning performance, which is counter-intuitive!

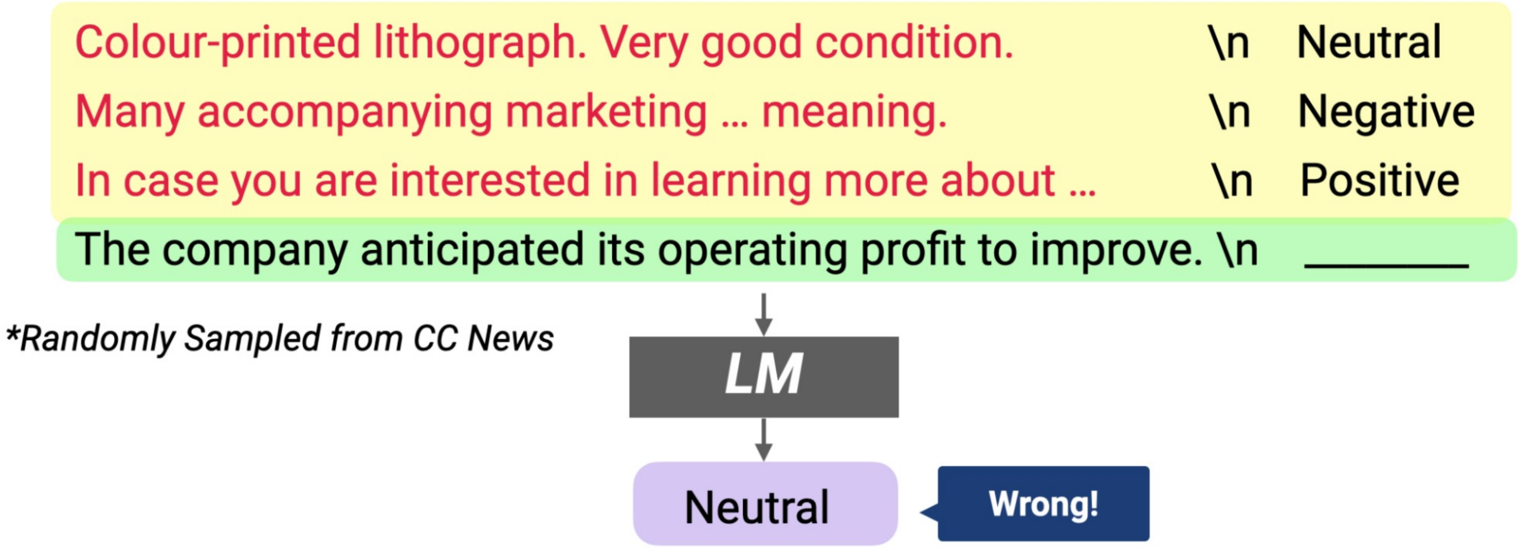




Comparison between no-examples (blue), examples with ground truth outputs (yellow) and examples with random outputs (random)

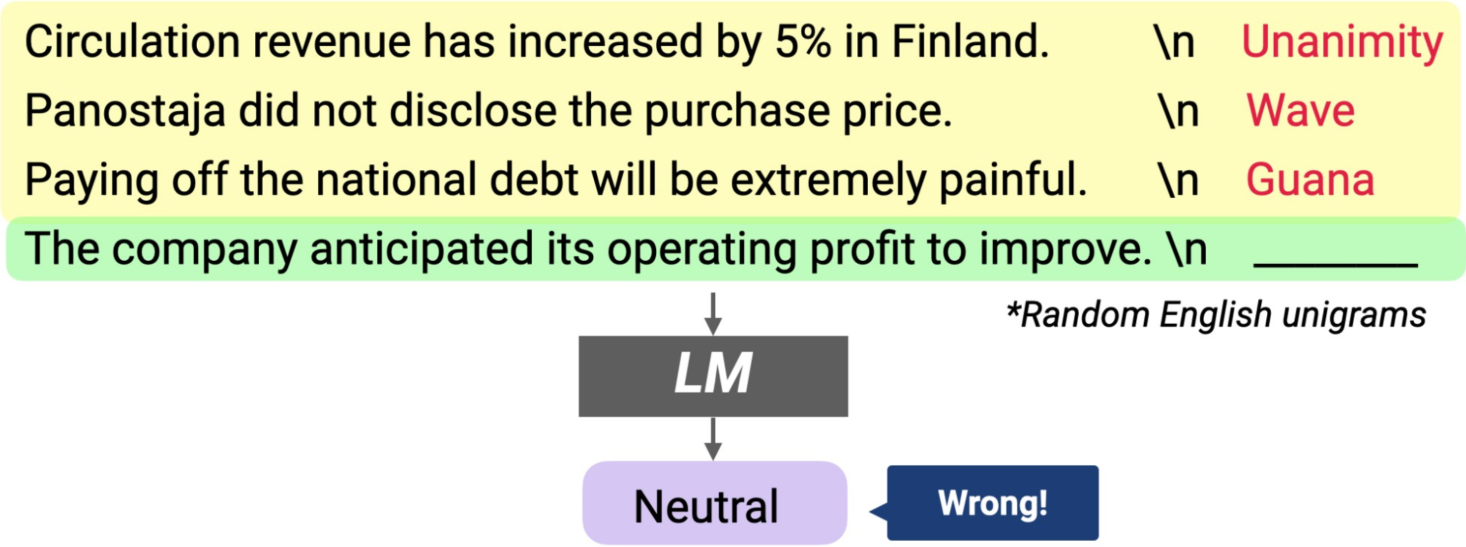
Which aspects of the prompt are most important for in-context learning? *Continued*

- **The input distribution matters:** when the inputs in the prompt are replaced with random inputs from an external corpus (the CC News corpus), model performance significantly drops. This indicates conditioning on the correct input distribution is important.



Which aspects of the prompt are most important for in-context learning? *Continued*

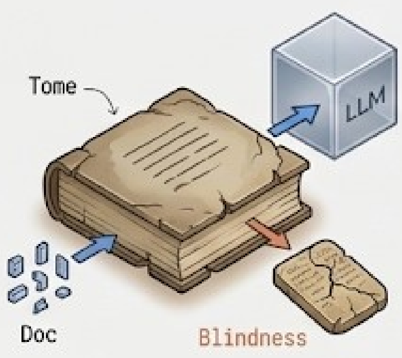
- **The output space matters:** when the outputs in the examples are replaced with random English unigrams, model performance significantly drops. This indicates conditioning on the correct output space is important.



Insights

- We can theoretically view in-context learning as Bayesian inference of a latent concept conditioned on the prompt, and this capability comes from structure (long-term coherence) in the pretraining data.
  - in-context learning is mainly about locating latent concepts learned during pretraining.
  - If terms in a particular instance are exposed many times in the pretraining data, the model is likely to know better about the distribution of the inputs.
- The in-context learning still works well when outputs in the prompt are replaced with random outputs.
  - While random outputs add noise and remove input-output mapping information, other components (input distribution, output distribution, format) still provide evidence for Bayesian inference.
- Specifying the task description can improve Bayesian inference by providing explicit observations of the latent prompt concept.

From Static Retrieval to Introspection (2020–2024)

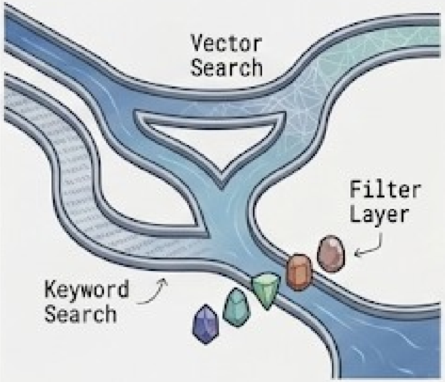


Naive RAG (The Baseline)

The Issue: Knowledge Cutoff. Models are blind to private data; retraining is costly.

The Innovation: Linear 'Search & Paste.' Vectorize docs, retrieve top-k, feed to LLM.

Critical Flaw: Blindness. Assumes retrieved docs are relevant; no quality verification.

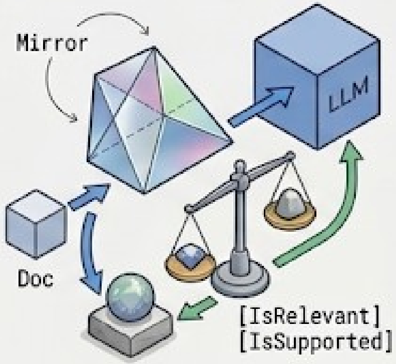


Advanced / Hybrid RAG

The Issue: 'Garbage in. Garbage Out.' Vague queries and missed keywords cause failure.

The Innovation: Pre & Post Optimization.

- **Pre-Retrieval:** Query Expansion & HyDE (Hypothetical Embeddings).
- **Retrieval:** Hybrid Search (Vector + BM25).
- **Post-Retrieval:** Cross-Encoder Reranking.



Self-RAG (The Critic)

The Issue: Blind Trust. Models hallucinate even with irrelevant docs.

The Innovation: Metacognition.

- **Reflection Tokens:** Outputs tags like `[IsRelevant]`, `[IsSupported]`.
- Critiques utility *before* answering.

CRAG (Corrective RAG)

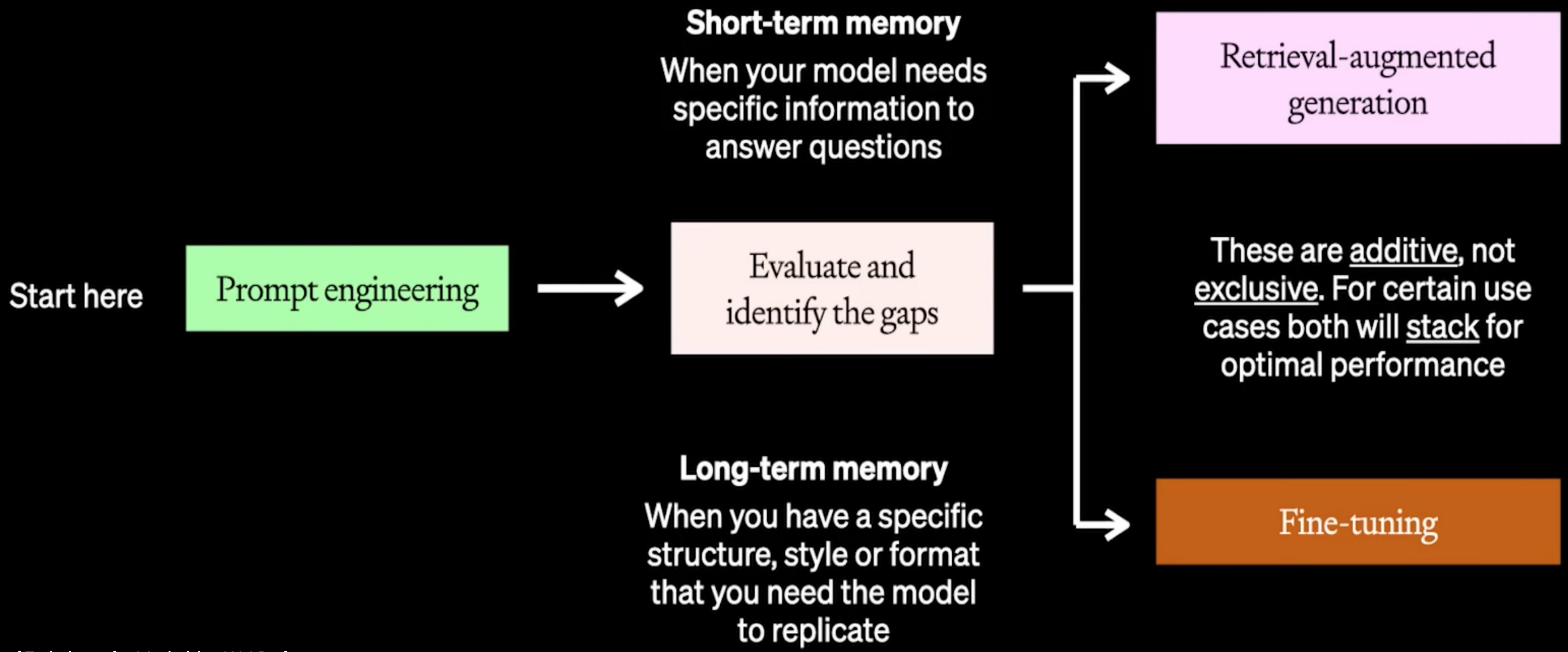
The Issue: Retrieval Failure. Internal DB lacks the answer.

The Innovation: The 'Retrieval Evaluator.'

- Scores retrieved docs.
- **High Score:** Use docs.
- **Low Score:** Trigger Web Search fallback.

2020 → 2022 → 2023 → Early 2024 →

RAG vs. Fine-tune



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